

CS 455 / CS 555 — Term Project Announcement

Large Language Models — Spring 2025/2026

Dear students,

It's time for your term project. In this project, I expect you to apply the LLM concepts, techniques, and tools we've covered throughout the semester to a real problem. Please read this announcement from start to finish, it answers most questions you'll have.

1. Project Philosophy

This project is not a formality for grading. The goal is to deepen what you've learned or will be learning this semester (transformers, fine-tuning, RAG, evaluation, prompting, agents etc.) by working hands-on on a system or research question that genuinely interests you. A good project is one where you learn by experimenting, discovering what LLMs can and cannot do in your chosen domain.

2. Group Structure

- **Group size: 2 or 3 students.** Single-person groups are not accepted. Groups of 4 or more are not accepted.
- Define your division of labor clearly within the group.
- **Finding a group:** If you don't have a group yet, use the SUCourse forum titled "*Find a Project Group / Project Group Formation Forum*" to find teammates. You may introduce yourself, mention your interests, preferred project ideas, and availability. Once your group is complete, update your post or reply to indicate that your group has been formed. Don't leave this to the last day.

3. Project Categories

You can work in any of the following categories (you may work on multiple categories at the same time):

(A) Application / System Building. Domain-specific RAG systems, LLM-powered tools, multi-agent systems, etc.

(B) Turkish NLP. Benchmarks for Turkish, fine-tuning Turkish models for specific tasks, low-resource techniques, etc.

(C) Evaluation / Analysis. LLM behavior analysis, bias, hallucination, robustness studies, prompt engineering comparisons.

(D) Efficiency / Optimization. Quantization, distillation, LoRA experiments, inference optimization.

(E) Research-oriented (especially for CS 555). Novel technique experiments, paper reproduction with extensions.

You are not limited to these categories. If you have a different idea, justify it in your proposal.

4. CS 455 Expectations

- A working system or analysis with clear results
- Appropriate evaluation with suitable metrics
- Final report (~8–10 pages) + code + short demo/presentation
- Honest error analysis: write what didn't work, not just what worked
- Reproducibility: code must be runnable, README must be clear

5. CS 555 Expectations (in addition to CS 455 requirements)

- Deeper literature review (5–10+ relevant papers)
- More sophisticated evaluation: baseline comparison, ablation study, a meaningful research question
- Final report (~12–15 pages, in academic format)
- Clear discussion of limitations and future work
- Where possible, a novel contribution (a new approach, dataset, or finding)

6. Optional: Paper Track

An optional track exists for groups (from either CS 455 or CS 555) who are interested in extending their project into a publishable research paper. This is **not required**, you can earn the highest grade (100) in the standard CS 455 or CS 555 track without applying.

Teams pursuing the Paper Track should:

- Explicitly state this intent in the proposal
- Be prepared to work closely with me on a bi-weekly basis
- Commit to continuing work into the summer
- Aim for publishable-quality results. Expectations are significantly higher than the standard course requirements

I will personally review Paper Track applications and select a small number of groups (likely 2–4) that I can supervise. Selection is based on the originality and feasibility of the proposal, the team's preparation, and the fit with my research capacity this term. **Not being selected is not a failure**, it simply reflects the limits of my capacity. Groups not selected continue in the standard CS 455 or CS 555 track without any penalty.

7. Compute & Resources

- **Free options:** Google Colab (free tier), Kaggle Notebooks (30h/week GPU), and free credits that some platforms (e.g., Google) offer to students.
- **Open-source / local models:** HuggingFace models and local inference are fully supported and recommended where appropriate.
- **Paid APIs (OpenAI, Anthropic, etc.):** You may use them, but **any cost is your own responsibility**.

- Choose your project scope realistically based on the compute available to you. "We couldn't run the model" is not an acceptable excuse for a failed project.

8. Data

- Use publicly available datasets where possible (HuggingFace Hub, Kaggle, academic dataset releases).
- **Web scraping:** If you scrape data, you are responsible for following the source's Terms of Service and applicable ethical and legal guidelines.
- **Private / institutional data:** If you use private data from a company, hospital, or any institution, you must obtain proper permission and follow all ethical and legal requirements (data protection, anonymization, etc.). Document this in your proposal and final report.
- Document your data source and any preprocessing in both the proposal and the final report.

9. Proposal → Format & Submission

Each group submits one proposal by the deadline. Only one group member should upload, multiple uploads from the same group will not be accepted.

Required Sections (in this order)

Section 1 — Group Members

List all group members in this exact format:

Name_Surname_of_student1, StudentID_of_student1

Name_Surname_of_student2, StudentID_of_student2

Name_Surname_of_student3, StudentID_of_student3

Section 2 — Track

State explicitly which one of the following applies (only one):

- CS 455 Project
- CS 555 Project
- Paper Track Application (open to both CS 455 and CS 555 groups) — optional

Section 3 — Project Proposal

- **Problem definition** (1 paragraph)
- **Motivation:** why is this important, why is it interesting (1 paragraph)
- **Approach:** what you'll do, which models/techniques you'll use (1–2 paragraphs)
- **Data:** what you'll use, source if any (1 paragraph)
- **Evaluation plan:** how you'll measure success (1 paragraph)
- **Timeline:** week-by-week plan
- **Risks and contingency plan:** what could go wrong, what's plan B
- **Compute resources:** what you'll use (Colab, Kaggle, Google Cloud Platform etc.)

Format & Filename

- **Format:** PDF only.
- **Length:** At most 6 pages.
- **Filename:** CS455or555_Proposal_Surname1_Surname2.pdf or CS455or555_Proposal_Surname1_Surname2_Surname3.pdf
- **Submission platform:** SUCourse only. No email submissions, no other platforms.

10. Important Dates

Milestone	Date
Proposal submission deadline	Sunday, May 10, 23:59
Proposal feedback to groups	Shortly after May 10 (TBA)
Final report + code submission	Sunday, June 7, 23:59
Project demos	After June 7 (details TBA)

You have more than a month from now to complete your project.

11. Working Before Proposal Approval

You may — and are encouraged to — start working on your project before the May 10 deadline. The feedback I (or the agent assistant — see Section 12) provide will typically be minor adjustments rather than major redirections.

However: if your proposal contains substantial errors, infeasible scope, or fundamental issues, the time lost is your responsibility. Plan accordingly.

12. Proposal Review Process

With approximately 110 students forming 35–55 groups, I cannot review every proposal individually in detail. The first-pass review will be performed by an AI assistant I have configured for this task. The assistant will:

- Evaluate your proposal against a defined rubric
- Provide structured written feedback (strengths, unclear points, scope concerns, etc.)
- Forward to me: (a) approved proposals as a summary, (b) proposals requiring my attention (Distinguished/Paper Track applicants, scope issues, etc.) in detail.

Important — please read carefully:

- **Final approval comes from me, not the assistant.** I am the instructor of record and the ultimate decision-maker for your project.
- **The AI assistant can make mistakes.** It is a tool to help me scale review across many groups, not a substitute for my judgment.

- **If the assistant's feedback conflicts with my view, my view prevails.** If you believe the assistant's feedback is wrong, unfair, or inconsistent with what I have said in class or office hours, contact me directly and I will review your proposal personally.
- This process does not mean you can submit a careless proposal. The assistant will be quite rigorous, and a poor proposal will get poor feedback regardless of who reviews it.

13. Track Changes During the Semester

- **Paper Track applicants who are not selected** will automatically continue in their stated CS 455 or CS 555 standard track, without any penalty.
- **You may not apply to the Paper Track after proposal approval**, this track requires commitment from the start.
- Selected Paper Track groups who later wish to step down to the standard track may do so with a brief notice to me. But this would be disappointing as I will spend extra time and not select other possible candidates. It may also cause a significant deduction in your grade.
- **If two groups propose substantially overlapping projects** (same problem, same method, same dataset), the group whose proposal was submitted earlier continues with the original idea; the other group(s) will be asked to differentiate their angle.

14. Final Deliverables (June 7, 23:59)

- **Final report** → PDF only, in the page-length format specified above for your course.
- **Code repository** → zip archive. Must include a clear README with setup and run instructions.
- Demos will be held after June 7. Demo format and scheduling details will be announced separately.

15. Late Submissions

No late submissions are accepted — for either the proposal or the final submission.

Plan around the deadlines. SUCourse will close at 23:59 on the relevant date. Technical issues at the last minute (slow internet, forgotten upload, etc.) are not valid excuses; submit early.

16. Academic Integrity

- You may use LLMs (Claude, GPT, etc.) as assistants, but you must clearly state how you used them in the report (which parts of the code, which sections of the writing, etc.).
- If you use code or ideas from other projects/papers, cite them properly.
- Report your results honestly. Writing "this didn't work" is always better than fabricating successful results. I respect honest negative results.
- Plagiarism (whether from other students, papers, or AI without disclosure) will be handled per university policy.

Good luck! I'm looking forward to seeing what you build.

Inanc Arin